



SYSTEM GUIDE · FOR REVIEW

TrotroLink

The digital system for fare compliance,
driver accountability and member finance

A plain-language guide to what the system is, who uses it, how every screen works,
which rules it enforces, and what GPRTU needs to provide to run it.

PREPARED FOR

Ghana Private Road Transport
Union (GPRTU)

VERSION

Review draft · 1 October 2026

SCREENS SHOWN

Demo build with sample data

START HERE

How to read this guide

TrotroLink is one system with three kinds of user. This guide follows them in turn, then explains the rules, the data, and what is still needed before a live launch.

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A note on the screens

All screenshots come from a demo build with sample data (vehicles, trips, fares, ratings and reports are examples). Where something is simulated, such as MTN MoMo payments or SMS, this guide says so. Phone screens are shown in the light theme unless a dark version helps; every screen exists in both.

1 · PURPOSE

What TrotroLink does for GPRTU

TrotroLink is not just a way for passengers to pay. It gives the union tools it has not had before: it records every fare, enforces the official price, knows which vehicles are registered, and lets the union act quickly on complaints and danger.

THE UNION'S PROBLEM	WHAT THE SYSTEM DOES	WHERE TO SEE IT
Fares are not respected	The price is set by the server from the union's fare table. The passenger sees the exact fare before paying and cannot change it. Overcharge reports record what the passenger was asked to pay; vehicles over 5% above the official fare are flagged.	Compliance · Fare tables
"Floating" drivers undercut the union	Only registered vehicle codes can take payments. Passengers see a "GPRTU Verified" badge, and can report a trotro with no sticker. The union can suspend a vehicle at once.	Registration · Compliance
No visibility of earnings	Every payment is logged with time, vehicle, route and amount. The union sees collections per vehicle and per terminal.	Overview · Transactions · Terminals
Terminal ticket leakage	Every trip books a small terminal fee to the terminal where its route starts. The terminal chairman can see it day by day.	Terminals
Fare changes cause confusion	The union schedules a new fare table with an effective time. From that moment the old fare stops working everywhere, drivers are told by SMS, and passengers see a notice.	Fare tables
Drivers cannot get credit	Each vehicle builds a verified record of fares collected. With the owner's consent the union can produce an income statement, by week or month, for a lender.	Statements
Poor conduct and danger	Passengers rate drivers and conductors. Accidents and careless driving reach the union live, with the passenger's location, and the union can reply within seconds.	Ratings · Live alerts · Disputes
Route discipline	A conductor can turn round for the return leg or change route, but only when no passengers are on board. Every change is logged for the union.	Registration

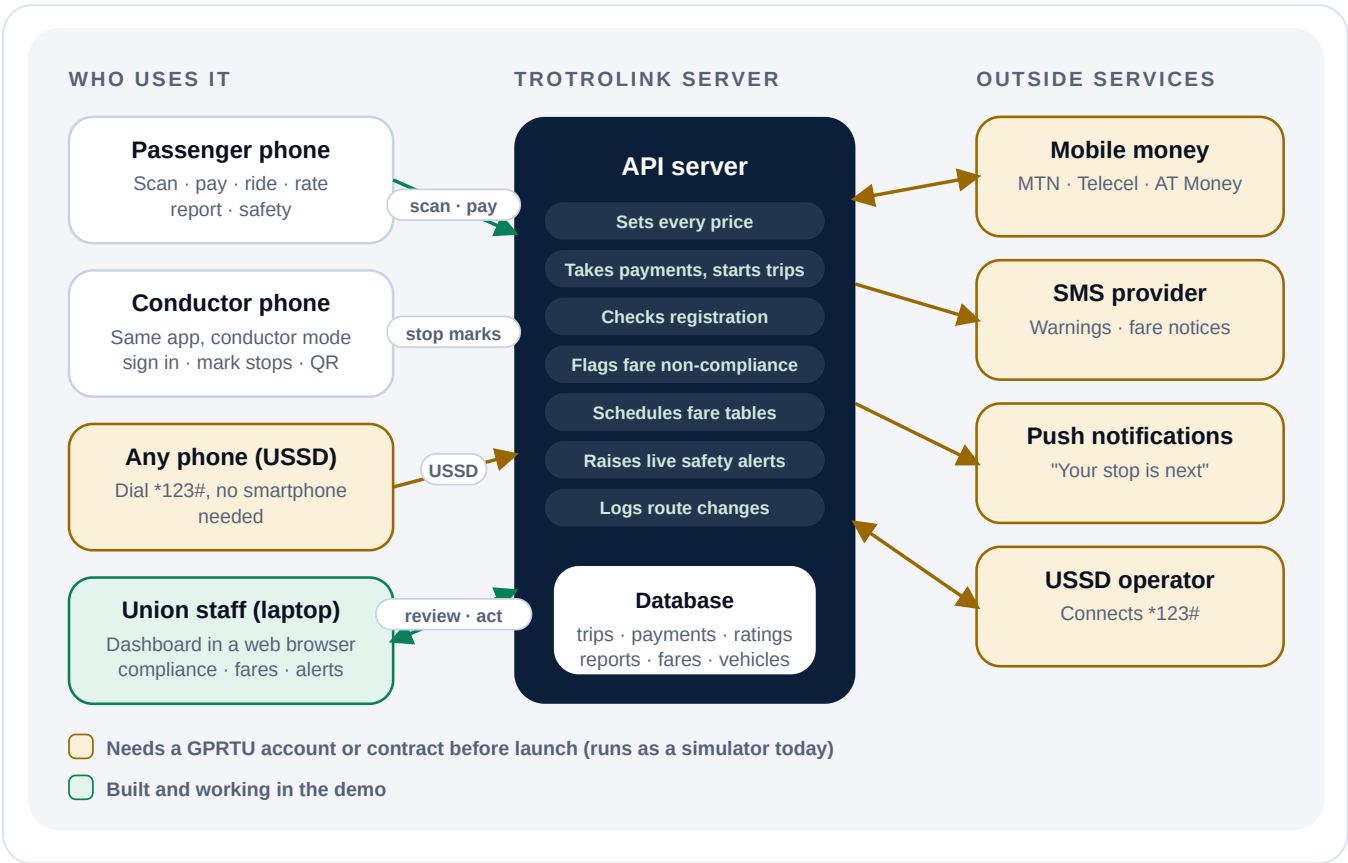
What the system cannot see

TrotroLink only knows about fares paid through the app. Cash paid to a conductor is not recorded, so fare enforcement relies on passenger reports as well as app payments, and statements cover only app fares. The more passengers pay through the app, the more complete the picture becomes.

2 · ARCHITECTURE

The system at a glance

Three things you touch, one server behind them, and four outside services. The server is the single source of truth: it decides prices, records every payment, and keeps the union's rules.



One app, two roles

Passengers and conductors use the same TrotroLink app. A conductor switches role in Profile and signs in with a vehicle code and PIN. Drivers do not need a separate app: the union reaches them by SMS.

The dashboard is a web page

Union staff open it in any browser on a laptop, enter the union key, and see everything. It refreshes live alerts every 5 seconds. It works in light and dark.

Works without a server too

For demonstrations and training, the app can run a complete offline demo inside the phone. It starts with a trip already under way so every screen can be shown.

3 · PEOPLE

Who uses TrotroLink

Each person sees only what they need. Nothing about the union's enforcement is visible to passengers, and passengers are never asked to create an account.

Passenger**Rides and pays**

- No sign-up: the phone is an anonymous guest
- Scans the QR sticker or types the short code
- Pays by MTN MoMo from their own phone
- Sees the live trip and ETA
- Rates the driver and conductor
- Reports problems, accidents, unregistered vehicles

Conductor**Runs the vehicle**

- Signs in with vehicle code and a 4-digit PIN
- Marks each stop as the trotro reaches it
- Sees paid passengers on board
- Prints or shares the QR sticker
- Turns round for the return leg, or changes route (when empty)
- Reports a passenger who did not pay

Union staff**Oversees and acts**

- Opens the dashboard in a browser
- Watches fares, ratings, collections
- Answers live safety alerts
- Warns, suspends and reinstates vehicles
- Publishes fare changes
- Issues income statements with owner consent

Drivers and owners

They do not use the app, but they are affected: the union sends them warnings and fare notices by SMS, and an owner's consent is needed before a statement of their vehicle's income is shared.

Visitors to Ghana

A built-in "Visiting Ghana?" guide explains trotros, MoMo, how prices work and how to stay safe, in plain language. Passengers can use the app in seven languages.

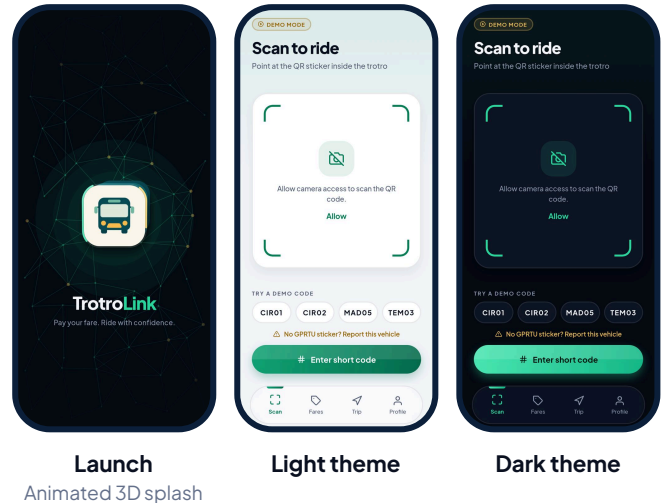
3 · DESIGN

The look and feel

The design aims at a premium, calm product that a first-time passenger can use in three taps and a union officer can read at a glance.

Principles

- **Three taps to ride.** Scan, choose your stop, pay.
- **One main action per screen.** A single green button.
- **Nothing hidden.** The exact fare is shown before paying; the union's verification is shown as a badge.
- **Safe by default.** Danger reports are one tap from the trip screen.
- **Works for everyone.** Seven languages; light and dark; large touch targets; labels for screen readers; haptic feedback on key moments.
- **Forgiving.** Poor network? Actions are queued and sent when the phone is back online.



Colour and type



Emerald is the action colour, gold is the highlight, deep navy and ink carry the dark theme. The font is **Plus Jakarta Sans**, chosen for clear numbers and the ¢ sign.

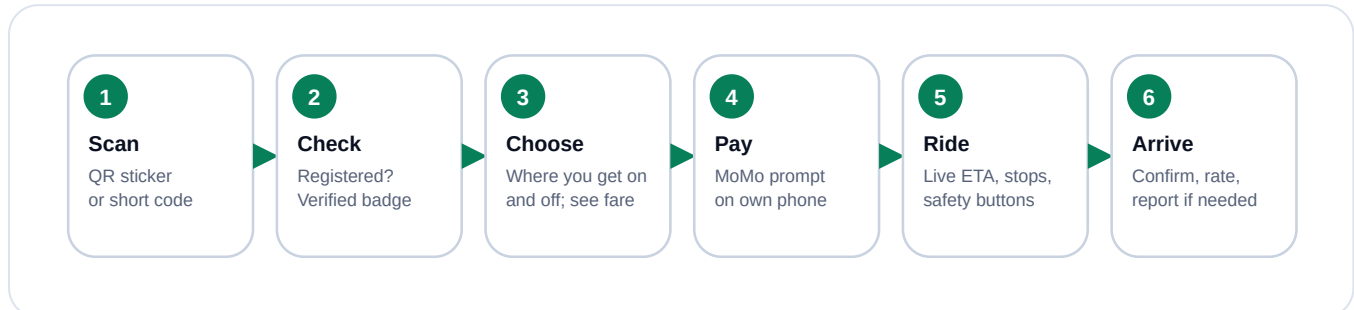
Languages

English, Twi, Ewe, German, Russian, Dutch and Chinese. The Twi and Ewe wording is a first version and should be reviewed by native speakers before launch. Conductor screens are in English.

4 · PASSENGER APP

A passenger's journey

From the sticker on the dashboard to a rating at the destination, in six steps. Every step produces a record the union can use.

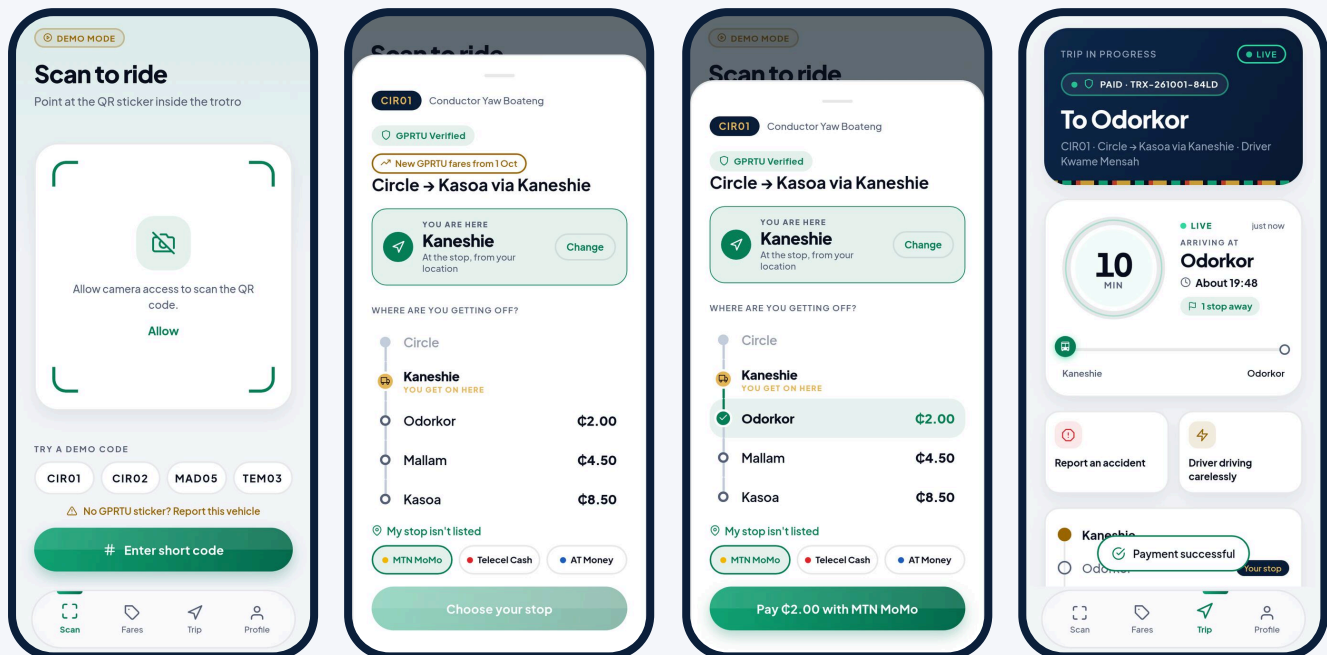


STEP	WHAT THE PASSENGER SEES	WHAT IT MEANS FOR GPRTU
1 · Scan	Camera with a viewfinder, or "Enter short code". Codes like C1R01 are printed under the QR sticker.	Only codes on the union's register are accepted. An unknown code offers "Report this vehicle to GPRTU".
2 · Check	The vehicle, conductor, route, a green "GPRTU Verified" badge, and any recent fare-change notice.	Passengers learn to look for the badge, which pushes floating drivers to register.
3 · Choose	The app finds the stop they are standing at from their location ("You are here"), so they only pick where they are going. Each stop on the route line shows its price.	Short hops are priced exactly. The price comes from the union's fare table, never from the phone.
4 · Pay	A MoMo request is sent to their phone; they approve with their PIN. The trip starts only when payment is confirmed.	Every trip is paid and logged before it begins: time, vehicle, route, amount.
5 · Ride	A live ETA, the route with a moving bus, and two safety buttons. Alerts when the stop is next.	If the trotro goes past the stop, the passenger is offered the extra fare; it is charged automatically after 60 seconds.
6 · Arrive	"You've arrived!": rate the driver and the conductor, with an optional comment.	Ratings build each driver's record and the Driver of the Day leaderboard.

4 · PASSENGER APP

Scan, choose, pay

The first half of the journey. The stop sheet is where the union's rules become visible to the passenger.



1 · Scan to ride

Viewfinder; demo codes for training; link to report a vehicle with no sticker

2 · Vehicle check

GPRTU Verified badge and a fare-change notice

3 · Stops, price, wallet

Getting on and off, exact fare, and which wallet pays

4 · Approve on phone

Wallet request (MTN, Telecel or AT); trip starts only after confirmation

If the vehicle is not allowed

A **suspended** vehicle shows a red "suspended by GPRTU" notice and cannot be paid. An **unknown code** is refused and the passenger is offered "Report this vehicle to GPRTU", which sends the code, a note and the phone's location to the union.

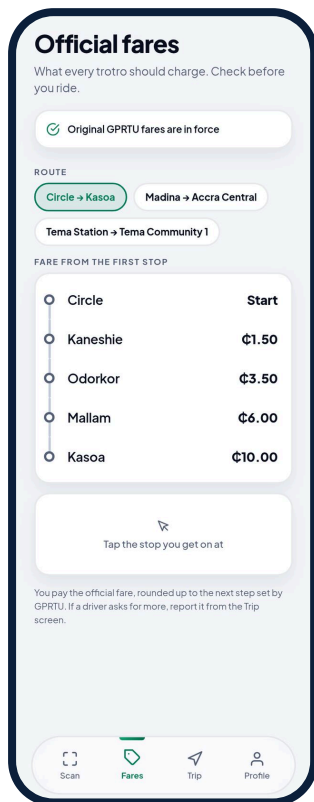
Prices, in plain terms

The fare between two stops comes from the union's table. The union can also set a specific price for a short hop, and decide how prices round (whole cedis today). If a trotro is already on the road, the sheet starts at the stop where it is now.

4 · PASSENGER APP

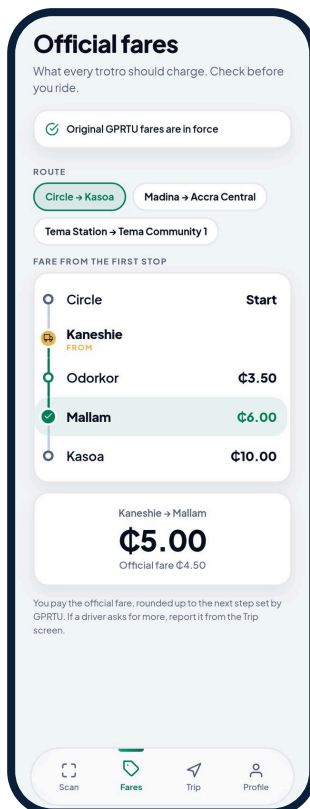
Fares and payment wallets

Passengers can check what a ride should cost before they scan, and choose the mobile-money wallet they pay from.



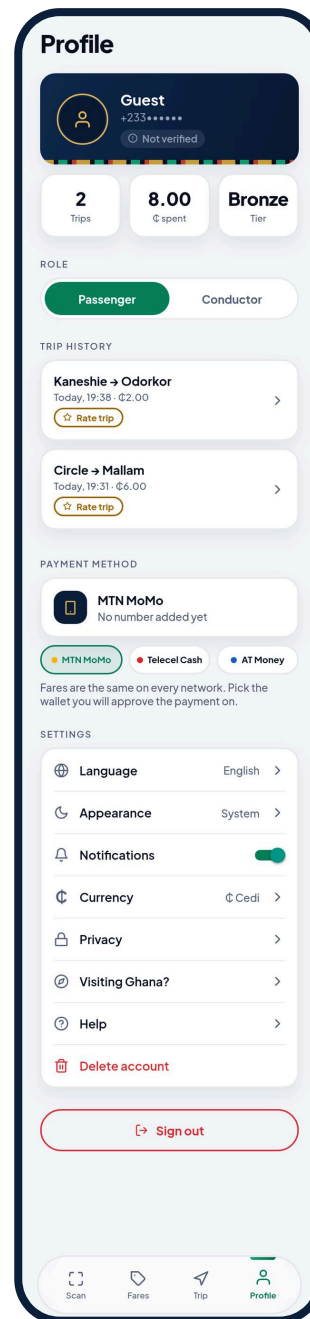
Fares tab

Official fares for every route, and the date they took effect



Short-hop check

Tap two stops to see the price between them



Payment wallet

MTN MoMo, Telecel Cash or AT Money

Fares anyone can check

The Fares tab reads the same table the union publishes on the dashboard. It shows the date the current fares started and warns of a change that is announced but not yet in force. It needs no scan and no sign-in, so a passenger can also use it to spot an overcharge.

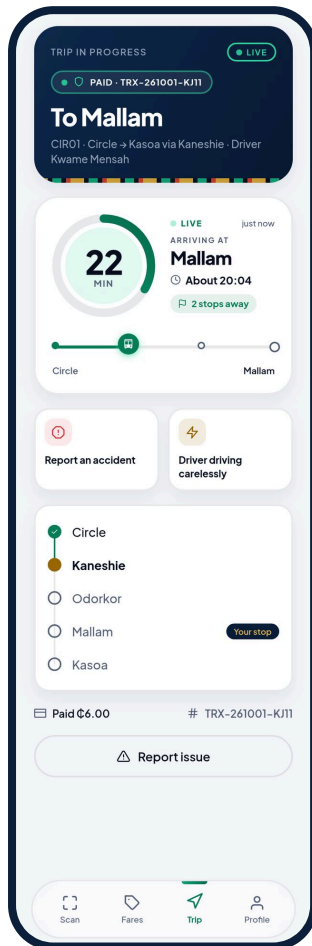
Which wallets work

The fare is the same whichever wallet pays. The app suggests a wallet from the phone number and remembers the choice. Only MTN is wired for live payments today; Telecel Cash and AT Money are shown in the demo and switch on when a provider that covers them is connected.

4 · PASSENGER APP

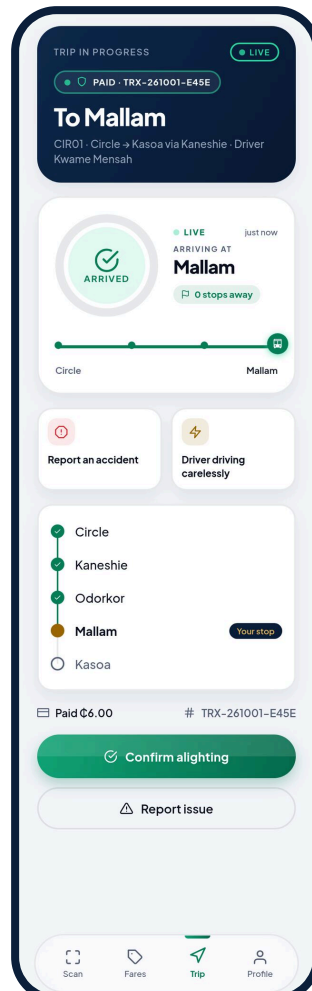
The live trip

Once paid, the passenger sees the journey unfold. The ETA is built to feel alive and trustworthy: it counts down, shows when the trotro should arrive, and shows where it is on the route.



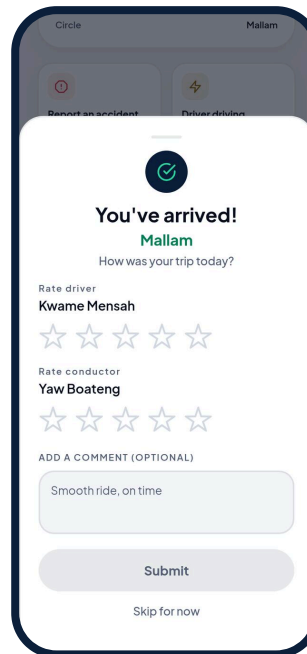
Live ETA

Progress ring, minutes, arrival time, moving bus



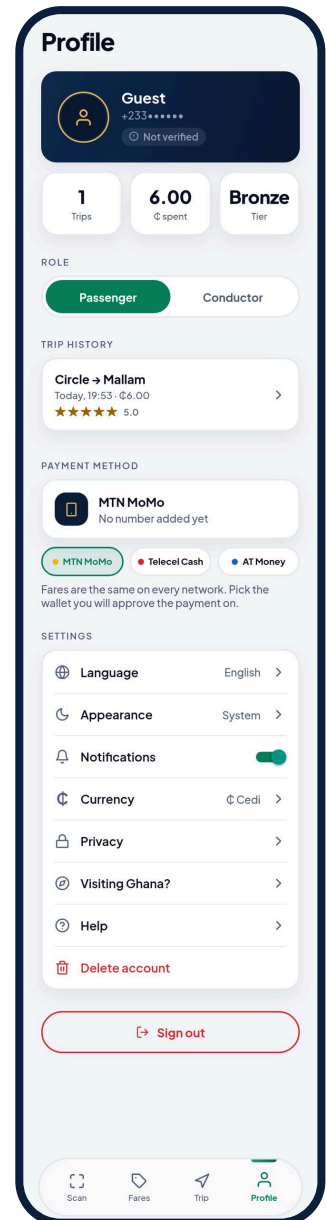
Arrival

At the stop: "Confirm alighting"



Rate the trip

Driver and conductor, 1–5 stars



History

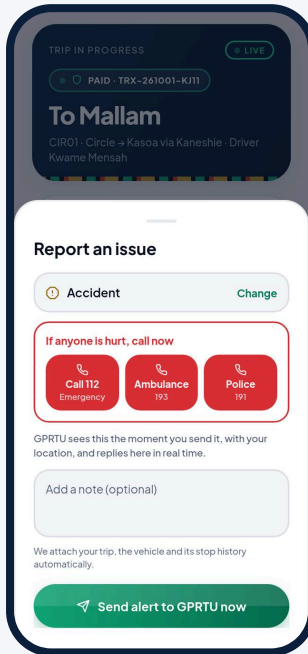
Past trips with ratings; "ride again"

- **Follows the road.** The passenger's own phone location moves the trip along, stop by stop. Nobody has to tap. Without location, the timetable ETA counts down.
- **Nudges at the right time.** "Your stop is next" and "You have arrived" alerts.
- **Overstay.** If the trotro passes the stop, the passenger can pay the difference or get off; if they do nothing the extra fare is charged after 60 seconds.
- **Mid-route boarding.** A passenger who gets on at Kaneshie sees a trip that starts at Kaneshie, not at the terminal.

4 · PASSENGER APP

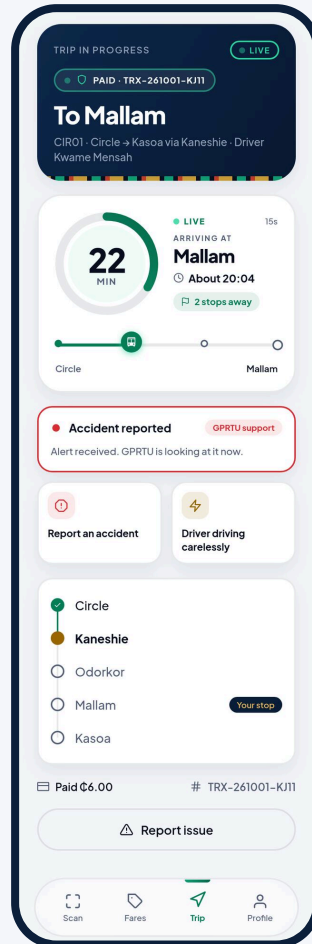
Safety and reporting

Two buttons sit under the ETA on every trip: **Report an accident** and **Driver driving carelessly**. They are live alerts, not forms to be read next week.



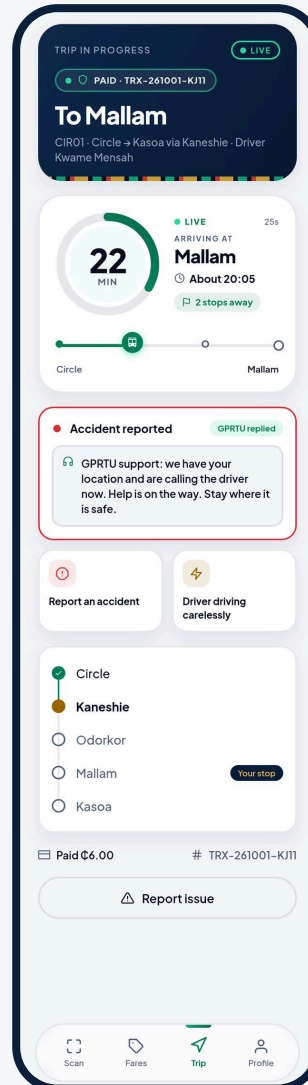
Accident

Call 112, ambulance 193, police 191 first; then alert GPRTU



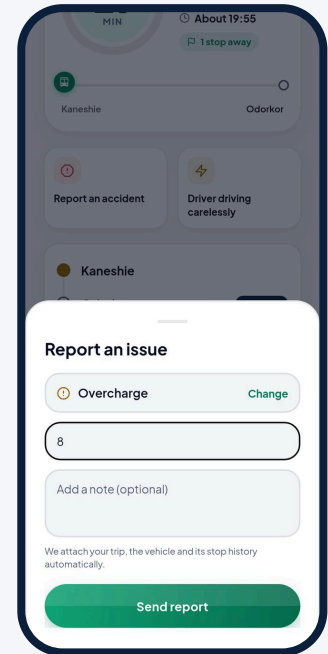
Alert sent

Appears on the dashboard at once, with the location



GPRTU replies

The union's answer shows in the trip, in real time



Overcharge

Asks what the passenger was asked to pay

How real-time support works

- The passenger sends the alert; the phone adds its location.
- It appears in **Live alerts** on the dashboard within seconds (the page checks every 5 seconds, and the sidebar count blinks red).
- Union staff can call the driver, see the location on a map, send a quick reply, and SMS the driver.
- The reply appears on the passenger's trip screen within seconds, with a tap to get their attention.

Other reports a passenger can make

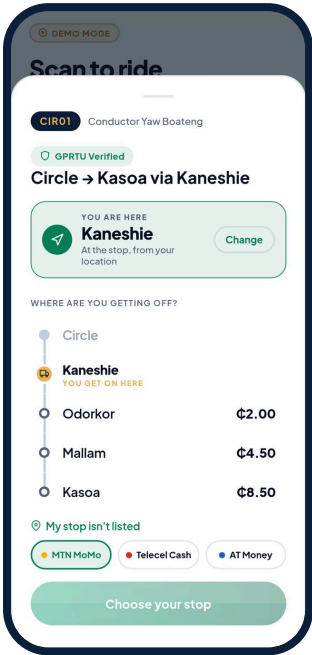
- **Overcharge** (with the amount asked)
- **Route deviation**
- **Safety concern**
- **Forced early alighting**
- **Unregistered vehicle** (from the Scan screen)

The system attaches the trip, vehicle, conductor and stop history automatically, so a report cannot be doctored afterwards.

4 · PASSENGER APP

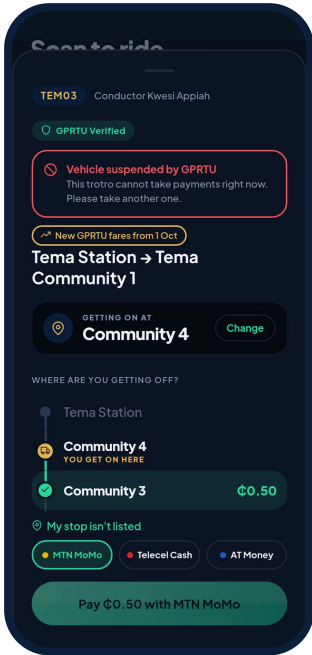
Short hops, return legs and other cases

Real trotro life is not always "terminal to terminal". The system handles the common variations.



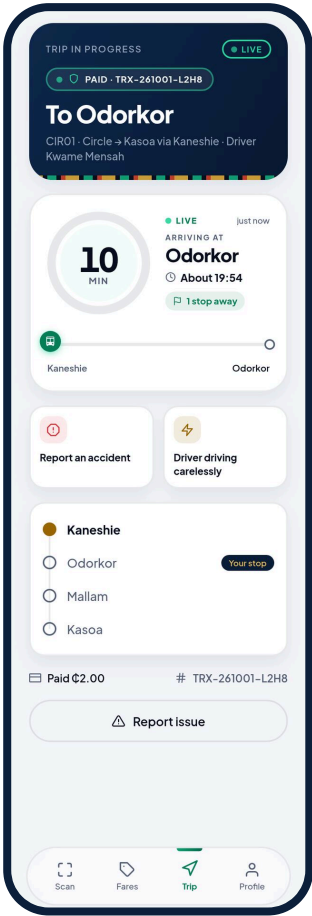
Getting on mid-route

Found from location; change it if wrong



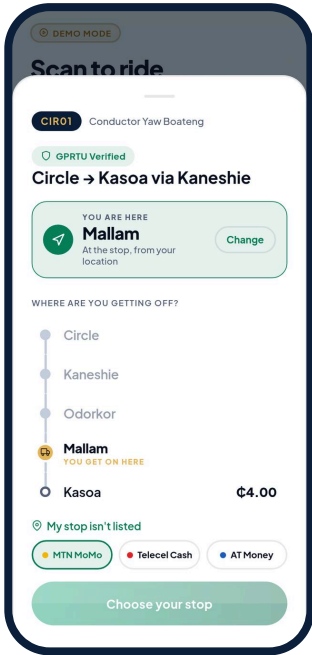
50 pesewa hop

Exact fare when the union sets 10p rounding



Trip starts there

ETA and stops begin at the boarding stop



Return leg

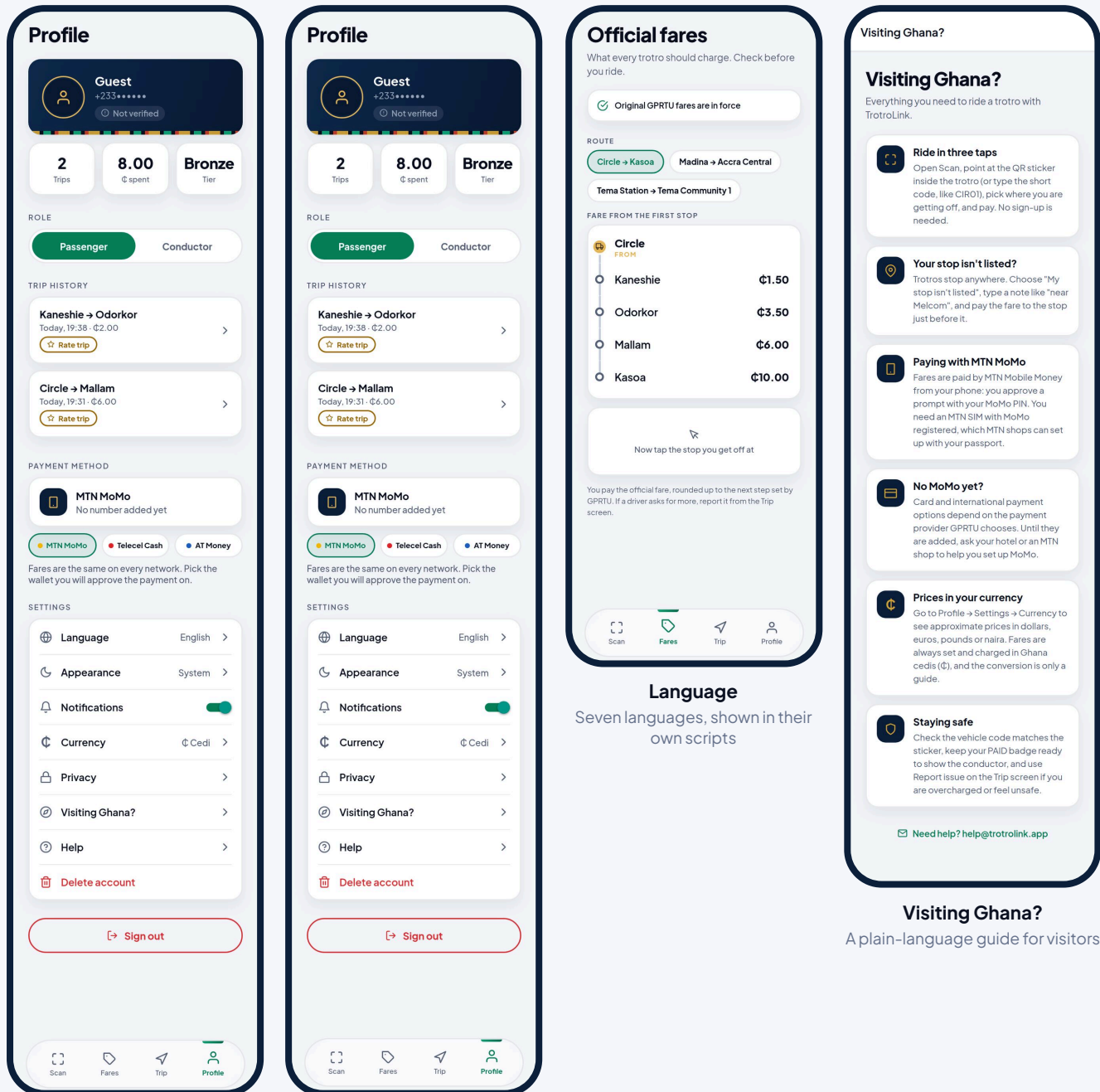
Kasoa → Circle: stops and fares reversed

SITUATION	WHAT HAPPENS
Stop is not on the list	The passenger taps "My stop isn't listed", types where (for example "near Melcom") and pays the fare to the stop before it. The union sees these notes under "Stop suggestions" and can add the stop to the route in one dialog.
No smartphone	The system includes a USSD menu (dial *123#): enter the vehicle code, pick a stop, pay by MoMo. It needs a USSD operator to be connected (see Launch checklist).
No network	Reports, ratings and stop marks are queued on the phone and sent when it reconnects. Payments always need the network.
A visitor to Ghana	The "Visiting Ghana?" guide explains how a ride works, MoMo, prices in familiar currencies (an approximate guide only; fares are always charged in cedis), and safety.

4 · PASSENGER APP

Profile and settings

A passenger's own space: trips, spending, payment method and preferences. The role switch to Conductor sits here too.



Profile

Trips, spend, tier, role switch

Settings

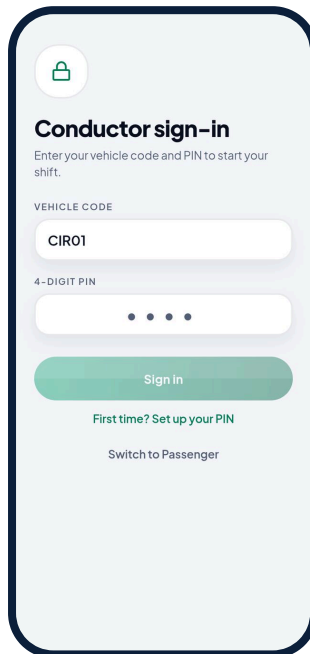
Language, appearance, notifications, currency, privacy, help

- **Trip history** with fares paid, ratings, and a "ride again" shortcut.
- **Appearance:** light, dark, or follow the phone.
- **Currency display:** approximate prices in other currencies as a guide; the charge is always in cedis.
- **Privacy and account deletion:** a passenger can read what is stored and delete their account and trip data.

5 · CONDUCTOR

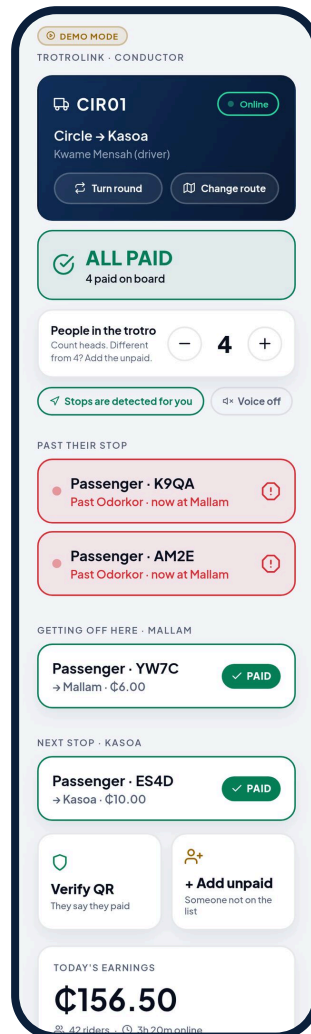
The conductor's day

Conductors use the same app in conductor mode. Conductors only watch. They do not mark stops or enter anything. The screen is built to be read at a glance, with big status badges, so that someone who does not read can still use it.



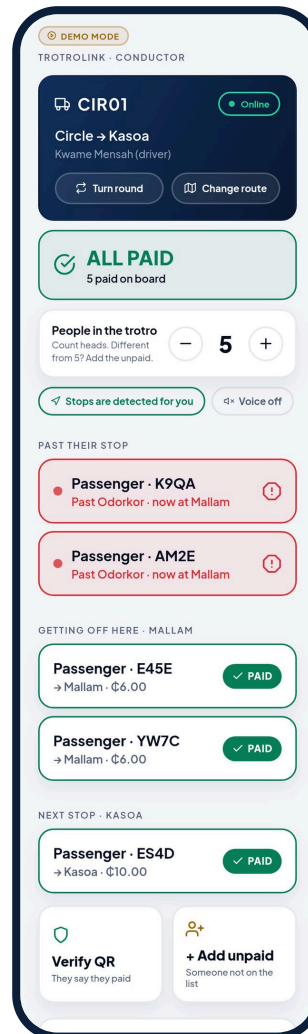
Sign in

Vehicle code and 4-digit PIN



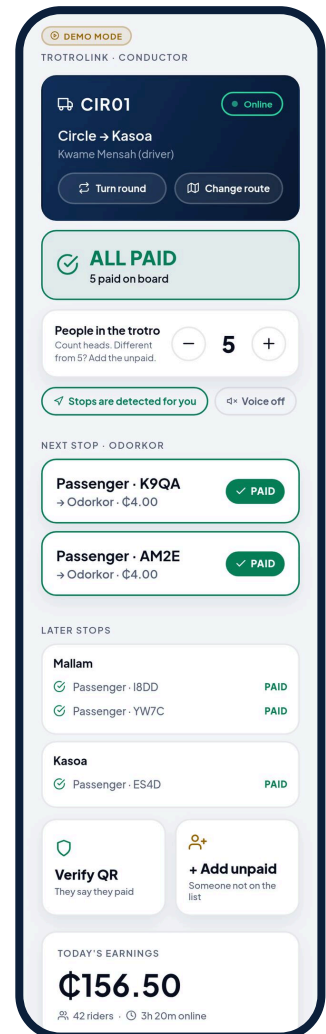
Today: the Big Board

Who is getting off, and who has paid



The board follows the road

Nobody marks stops; it updates by itself



Safe changes

Turning round is blocked while passengers are on board

Signing in

- First time: the conductor chooses a PIN for their vehicle (a setup code from the union can be required).
- Five wrong PINs lock the account for 15 minutes; a session lasts 12 hours.
- If a conductor forgets, only the union can reset the PIN.

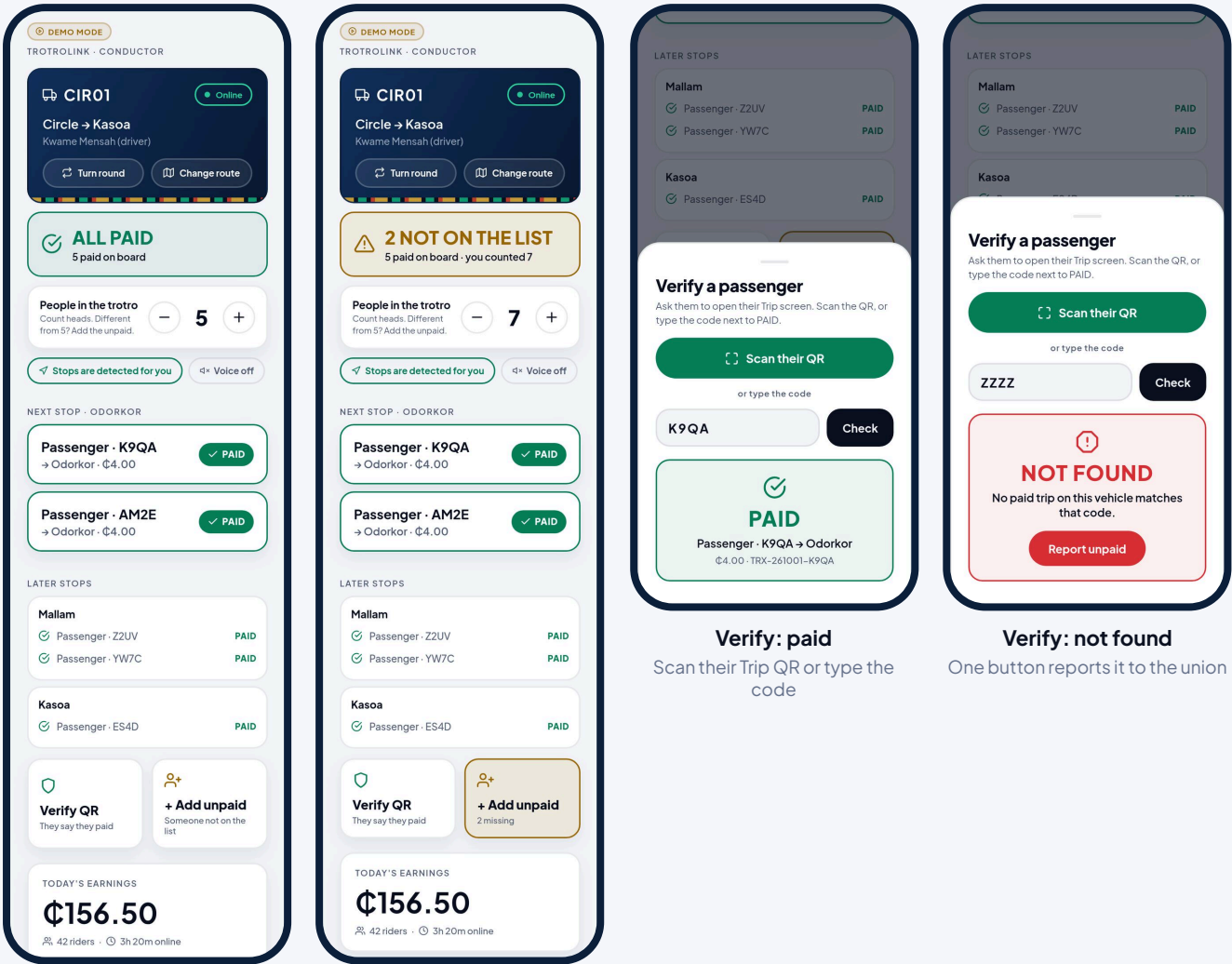
On the road

- Nothing to mark: stops are found from the phones on board, and passengers are told "your stop is next".
- Read who is getting off next, count heads, compare with the number on screen.
- Add an unpaid passenger, or verify a passenger who says they paid.
- **Turn round** for the return leg, or **Change route**. Both are refused with passengers on board and are logged for the union.

5 · CONDUCTOR

The Big Board

The conductor's main screen answers three questions without any reading: who is getting off next, has everyone paid, and does the head count match.



Who is getting off

Next stop first, paid badges, later stops grouped

Head-count check

7 people counted, 5 paid: 2 not on the list

Fewer taps

MOMENT	EFFORT
Boarding	Nothing: passengers pay themselves
Stop reached	Nothing: found from the phones on board
Someone says they paid	1 tap: Verify QR
Someone not on the list	1 tap: Add unpaid
End of shift	1 tap

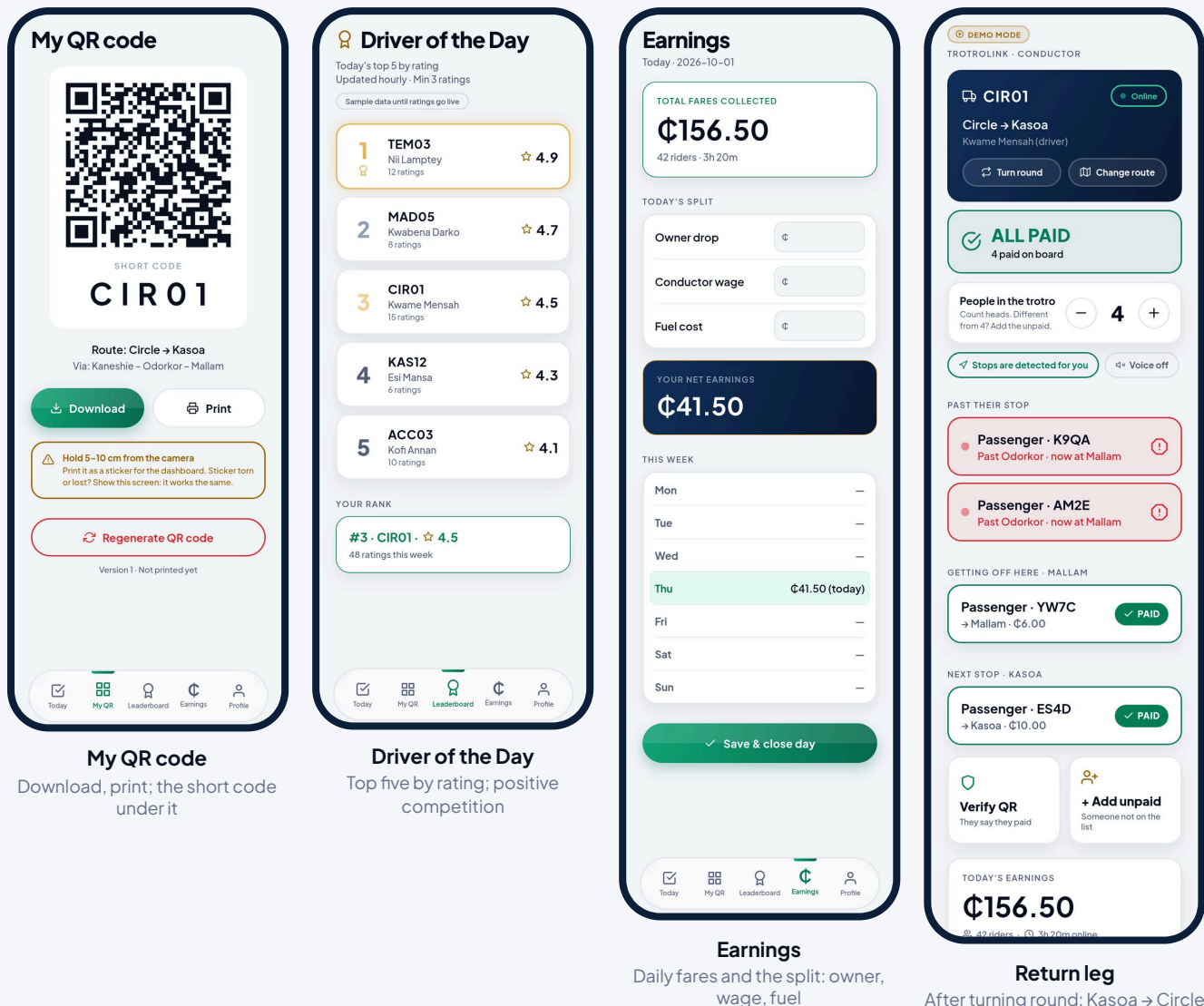
Good to know

- Printed sticker and phone QR both work. If the sticker is torn or lost, the My QR screen shows the same code.
- Passengers are anonymous guests, so the board shows a short trip code (for example "K9QA"), not a name.
- Voice readout is optional: it says the next stop and who is getting off.
- Turn round happens by itself when someone scans at the end of the line with nobody on board. A big button is there if it does not.

5 · CONDUCTOR

Sticker, ranking and earnings

Supporting screens that keep conductors engaged and give the vehicle its identity.



Note on sample figures

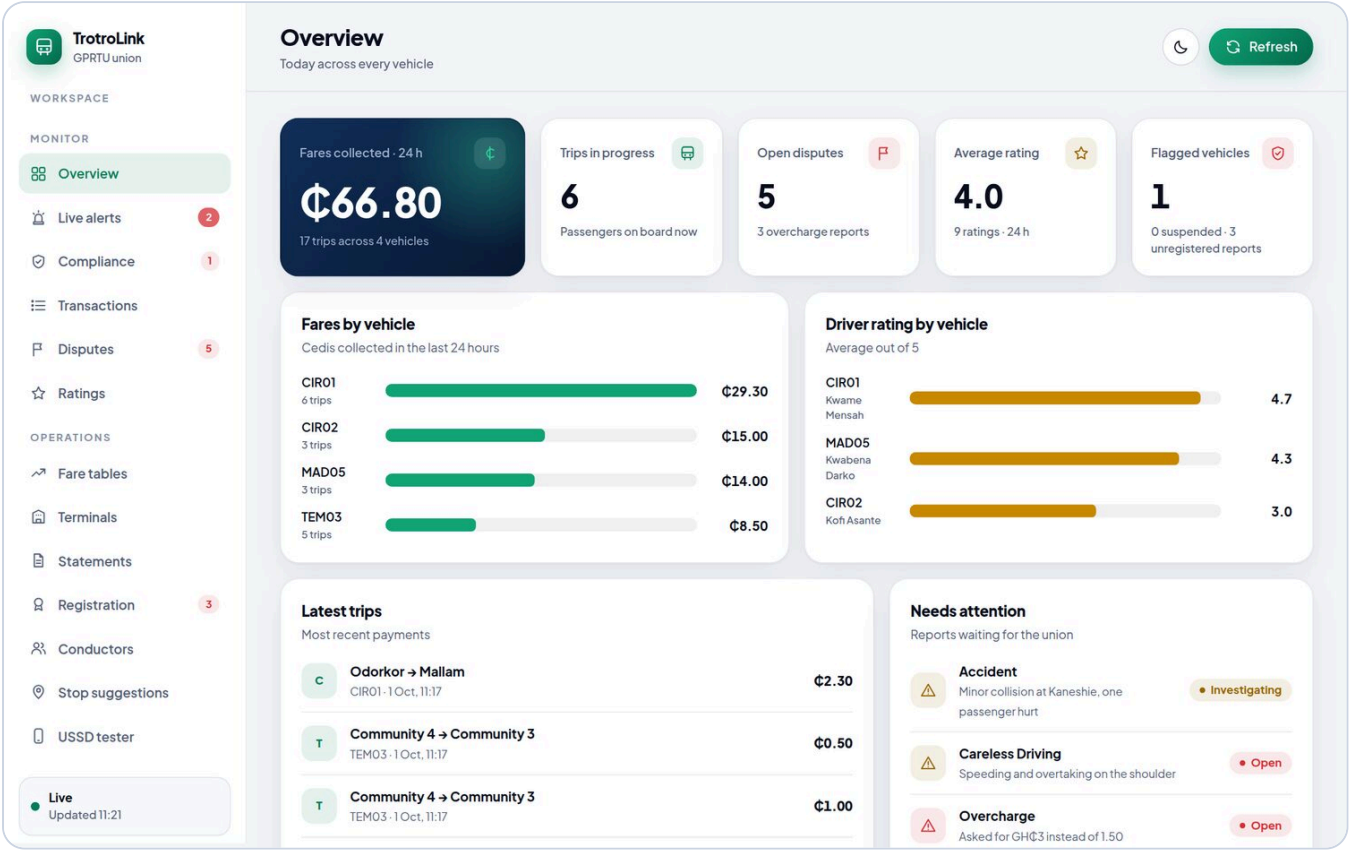
In the demo, the conductor's "Today's earnings", scan count and bonus progress are placeholder numbers. The daily split on the Earnings tab is entered by the conductor. These screens are designed, but their figures are not yet fed from live payments. This is listed under "What is live, simulated or still needed".

- **The QR sticker is the vehicle's identity.** It can be regenerated if a sticker is damaged or stolen; the short code is printed beneath it.
- **The leaderboard** uses passenger ratings, and the union can reward top performers.
- **Earnings** help conductors plan; they are separate from the union's verified statements, which come from app payments only.

6 · UNION DASHBOARD

The dashboard

Open it in a browser, enter the union key, and the whole picture is in one place. The sidebar groups the work into **Monitor** (what is happening) and **Operations** (things the union sets up or changes). Red counts show what needs attention.



PART OF THE PAGE	WHAT IT TELLS YOU
Fares collected · 24 h (large card)	Cedis paid through the app in the last day, and how many trips on how many vehicles.
Trips in progress / Open disputes / Flagged vehicles / Average rating	The four numbers to watch: who is riding now, what is waiting for a decision, which vehicles are in trouble, and passenger satisfaction.
Fares by vehicle · Driver rating by vehicle	Which vehicles earn most, and who passengers rate highest. Hover a bar for details.
Latest trips · Needs attention	The last payments, and reports waiting for the union.

The dashboard has a light/dark switch (top right) and works on a laptop or a large phone. Access is protected by a union key.

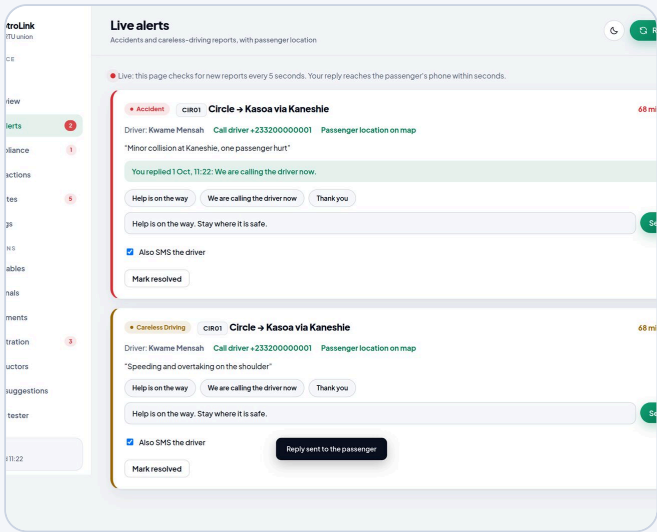
6 · UNION DASHBOARD

Live alerts and Compliance

Live alerts

Accidents and careless-driving reports from passengers appear here the moment they are sent, newest first. The page refreshes every 5 seconds, and the sidebar count blinks red.

- **See:** type, vehicle, route, driver, how long ago, passenger note.
- **Locate:** "Passenger location on map" opens a map at the phone's position.
- **Respond:** call the driver, send a quick reply to the passenger, SMS the driver, then mark resolved.



Compliance

One row per vehicle with its status, rating, overcharge reports, warnings, flags and a level (Clear, Watch, Warning, Critical). The three rules at the top are the union's thresholds.

TrotroLink
GPRTU union

WORKSPACE

MONITOR

Overview

Live alerts 2

Compliance 1

Transactions

Disputes 5

Ratings

OPERATIONS

Fare tables

Terminals

Statements

Registration 3

Conductors

Compliance

Fare enforcement, warnings and suspensions

Fare over official by >5%

Driver rating below 3.0 (min 3 ratings)

2 warnings in 90 days: suspension recommended

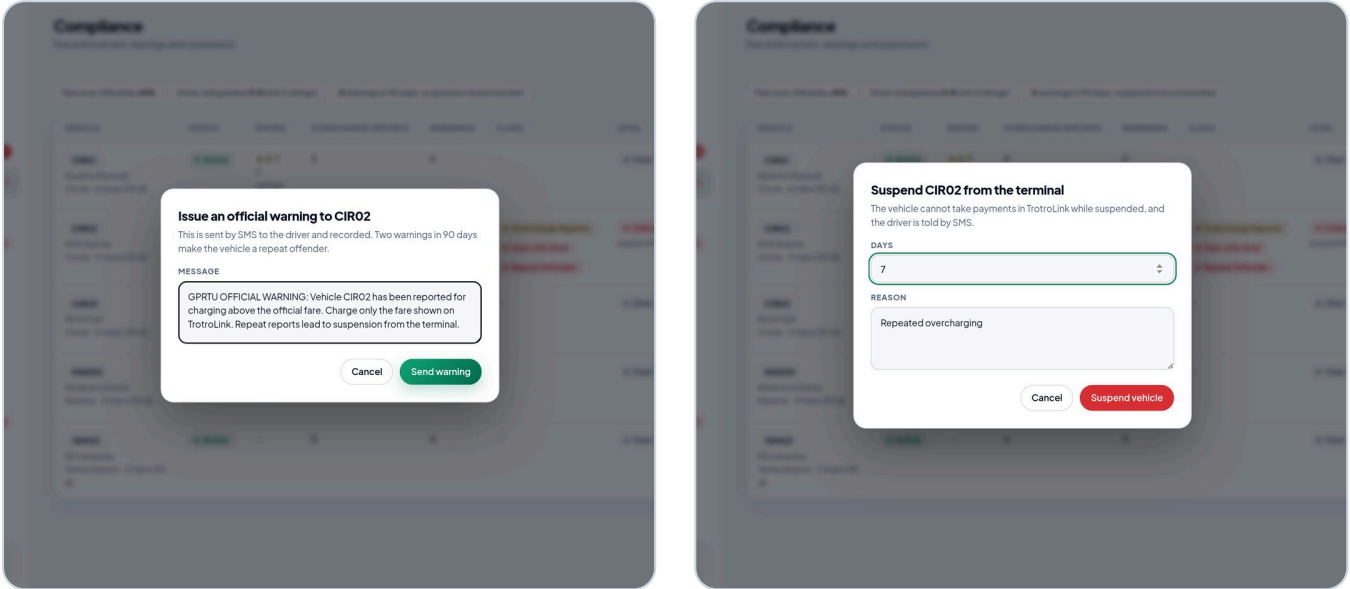
VEHICLE	STATUS	RATING	OVERCHARGE REPORTS	WARNINGS	FLAGS	LEVEL	ACTION
CIR01 Kwame Mensah Circle - 6 trips (30 d)	Active	★ 4.7 3 ratings	0	0	—	Clear	Issue warning Suspend
CIR02 Kofi Asante Circle - 3 trips (30 d)	Active	★ 3.0 3 ratings	3 asked +66.7% over	2	Overcharge Reports Fare >5% Over Repeat Offender	Critical suspend?	Issue warning Suspend
CIR03 Nii Armah Circle - 0 trips (30 d)	Active	—	0	0	—	Clear	Issue warning Suspend
MAD05 Kwabena Darko Madina - 3 trips (30 d)	Active	★ 4.3 3 ratings	0	0	—	Clear	Issue warning Suspend
TEM03 Nii Lamptey	Active	—	0	0	—	Clear	Issue warning Suspend

FLAG	RAISED WHEN	WHAT THE UNION CAN DO
Overcharge reports	A passenger reported an overcharge in the last 30 days.	Review the evidence in Disputes; issue a warning.
Fare >5% over	Passengers say they were asked, on average, more than 5% above the official fare.	Issue an SMS warning.
Rating <3.0	Driver rating is below 3.0 with at least 3 ratings.	Review; talk to the driver.
Repeat offender	Two warnings within 90 days. Suspension is recommended.	Suspend, with days and a reason.

6 · UNION DASHBOARD

Warnings, suspension and reinstatement

The union's authority, turned into two buttons with a record. Every action is logged, and a suspension takes effect immediately at every passenger's phone.



ACTION	WHAT HAPPENS	EFFECT
Issue warning	A pre-written, editable message is sent by SMS to the driver's phone and recorded.	Counts towards "repeat offender". (SMS is simulated until an SMS account is connected.)
Suspend	The union enters a number of days and a reason. The driver is told by SMS.	Passengers see "Vehicle suspended by GPRTU"; payments, including by USSD, are refused.
Reinstate	One tap.	The vehicle takes payments again.

Why this matters

Enforcement has no value if the vehicle can keep working. Because every payment goes through the server, a suspension is not advice: the vehicle genuinely cannot take app payments until the union reinstates it.

6 · UNION DASHBOARD

Fare tables

The union sets fares here. From the effective time, every price the app shows and accepts comes from the new table, so nobody can charge the old fare after the deadline.

TrotroLink
GPRTU union

WORKSPACE

MONITOR

Overview

Live alerts 2

Compliance 1

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Live
Updated 11:22

Fare tables

Schedule fare changes and tell every driver

Schedule a fare change

From the effective time, every price the app shows and accepts comes from the new table. The old fare stops working everywhere at once.

NAME

e.g. Fare review, October 2026

EFFECTIVE FROM

10/08/2026, 11:22 AM

HOW

% change

Fare list

CHANGE ALL FARES BY (%)

8

PASSENGERS' PRICES ROUND UP TO

10 pesewas (best for short hops)

With whole ceddis, a 50 pesewa hop costs ₦1.00. A smaller step charges the exact fare for short trips.

Preview change

Publish and tell drivers

Publishing sends an SMS to every driver (Hubtel) and a push to every conductor. Without SMS credentials it runs in simulator mode and sends nothing real.

Short hops

ROUTE	FROM	TO	WAS	NEW	PASSENGER PAYS
TEM-C1-01	Community 4	Community 3	₦0.50	₦0.50	₦0.50

Fare change history

NAME	EFFECTIVE	CHANGE	ROUNDS TO	TOLD DRIVERS	STATE
SAMPLE: +8% review (not official)	1 Dec. 00:00	+8%	₦1.00	5 drivers · 1 Oct. 09:45	Scheduled
GPRTU fares, October 2026 (sample)	1 Oct. 00:00	0%	₦1.00	not sent	In Force
SAMPLE: short hops + 10p rounding	1 Oct. 00:00	custom list	₦0.10	not sent	In Force

To publish a fare change

- Name it (for example "Fare review, October").
- Choose the effective date and time.
- Either enter a **% change** for all fares, or paste a **fare list**: one line per stop (*route, stop, fare*) or per short hop (*route, From > To, fare*).
- Choose how prices **round up**: whole ceddis (current), 50p, 10p or 5p.
- **Preview** every change, then **Publish and tell drivers**.

What the change does

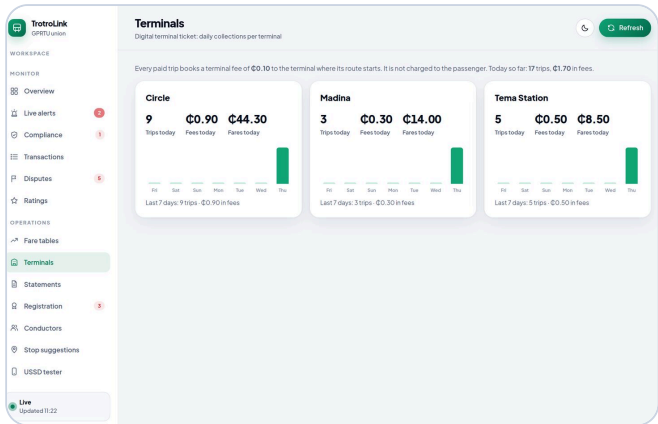
- Drivers get an SMS; conductors a push notification.
- Passengers see "New GPRTU fares from ..." for 14 days.
- The latest published table wins; older ones stay in the history.
- **Short-hop fares** let the union price a specific stop-to-stop trip exactly, even when it is not the usual difference.
- The fares in the demo are samples. GPRTU's official table replaces them.

6 · UNION DASHBOARD

Terminals and Statements

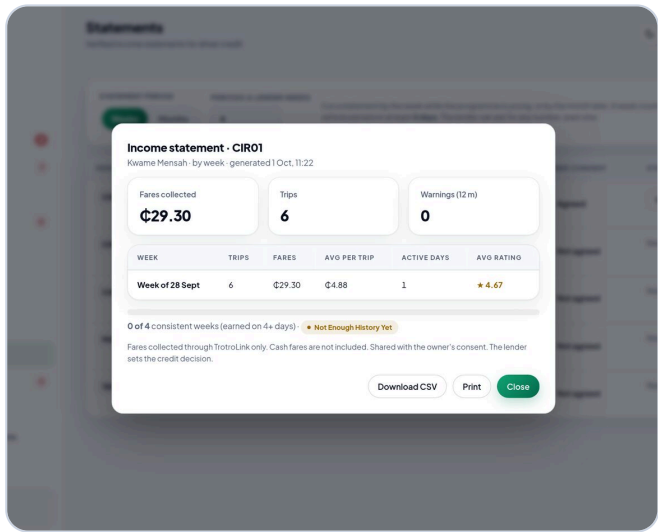
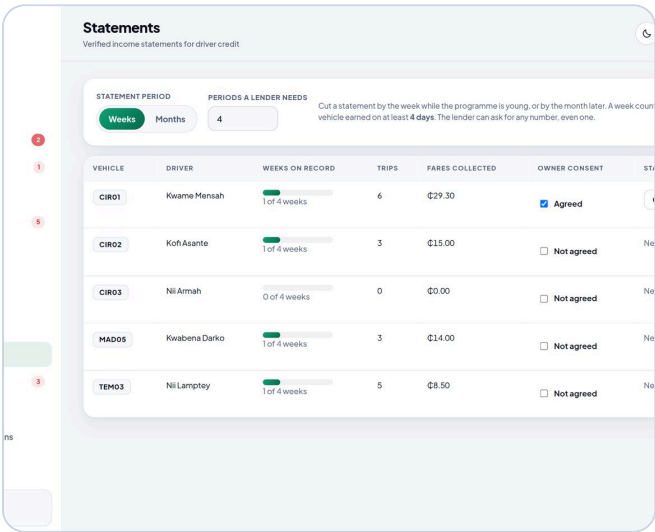
Terminals: the digital terminal ticket

Every paid trip books a small **terminal fee** (GH¢0.10 today, adjustable) to the terminal where its route starts. It is not charged to the passenger. Each terminal card shows today's trips, fees and fares, and a 7-day chart. The terminal chairman can see collections in real time, which reduces leakage from cash handling and gives proof of revenue for terminal improvement funds.



Statements: verified income for driver credit

A lender wants proof of steady earnings. TrotroLink can produce a statement per vehicle from the fares it collected, by **week or by month**, with the number of periods the lender needs. A statement is only produced when the owner's consent has been recorded. It shows trips, fares, average per trip, active days and average rating, and whether the vehicle meets the history requirement. It can be printed or downloaded as a CSV file.



The history requirement shown (for example 4 consecutive weeks, or 6 months) is a placeholder: the lender decides the real one. Cash fares are not included.

6 · UNION DASHBOARD

Registration and Conductors

Registration

Shows how many vehicles are registered (these carry the "GPRTU Verified" badge), how many are suspended, and the reports passengers have made of **unregistered vehicles** (code seen, note, map link, status). It also lists every **route and direction change** made by a conductor, with the vehicle, what changed and when. More than two route switches by one vehicle in a day is flagged as unusual.

TrotroLink
GPRTU union

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Registration

Registered vehicles and reports of unregistered ones

Registered vehicles
5
Show "GPRTU Verified" in the app

Suspended
0
Cannot take payments

Unregistered reports
3
Open, reported by passengers

Reports of unregistered vehicles

REPORTED	CODE SEEN	DETAILS	LOCATION	STATUS
1 Oct, 10:03	XYZ99	Red trotro near Kaneshie market, no sticker	—	open
1 Oct, 10:02	XYZ99	Red trotro near Kaneshie market, no sticker	—	open
1 Oct, 09:45	XYZ99	Red trotro at Kaneshie, no sticker	View on map	open

Route and direction changes

Conductors can turn a trotro round for the return leg or move it to another route, only when no passengers are on board. Every change is logged here. More than two route switches in a day is flagged.

WHEN	VEHICLE	CHANGE	FROM	TO	NOTE
1 Oct, 10:49	CIR01 Kwame Mensah	Route Changed	MAD-ACC-01 outbound	CIR-KSA-01 outbound	
1 Oct, 10:49	CIR01 Kwame Mensah	Route Changed	CIR-KSA-01 inbound	MAD-ACC-01 outbound	

Conductors

Each vehicle's conductor, driver, whether a PIN is set, last sign-in, and whether the account is locked. The union can **reset a PIN**, which ends the conductor's sessions; the conductor then sets a new one.

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PIN status and resets

Resetting a PIN ends the conductor's sessions. They then choose a new PIN with "First time? Set up your PIN".

VEHICLE	CONDUCTOR	DRIVER	PIN	LAST SIGN-IN	ACTION
CIR01	Yaw Boateng	Kwame Mensah	Set	1 Oct, 10:49	Reset PIN
CIR02	Kojo Owusu	Kofi Asante	Not Set Up	—	
CIR03	Ebo Quaye	Nii Armah	Not Set Up	—	

6 · UNION DASHBOARD

Transactions, Disputes, Ratings

Transactions

Every payment, newest first: time, trip reference, vehicle, route, **official fare**, **amount paid**, and whether the passenger's phone was near the stop when they got off. This is the audit trail.

TIME	TRIP	VEHICLE	ROUTE	OFFICIAL FARE	PAID	GPS AT ALIGHTING
1 Oct, 11:17	TRX-HOP-0004	CIR01	Odokor + Matam	€2.30	€2.30	—
1 Oct, 11:17	TRX-HOP-0003	TEM03	Community 4 + Community 3	€0.50	€0.50	—
1 Oct, 11:17	TRX-HOP-0002	TEM03	Community 4 + Community 3	€0.50	€1.00 rounded up	—
1 Oct, 10:03	TRX-241001-EIPD	CIR01	Circle + Odokor	€3.50	€4.00 rounded up	—
1 Oct, 10:02	TRX-241001-FHE1	CIR01	Circle + Odokor	€3.50	€4.00 rounded up	—
1 Oct, 08:20	TRX-241001-S012	TEM03	Community 4 + Community 1	€1.50	€2.00 rounded up	—
1 Oct, 08:20	TRX-241001-S011	TEM03	Tema Station + Community 1	€1.50	€3.00 rounded up	—
1 Oct, 08:20	TRX-241001-S010	TEM03	Tema Station + Community 2	€2.00	€2.00	—
1 Oct, 08:20	TRX-241001-S009	HAD05	Lagon + Accra Central	€4.50	€5.00 rounded up	—
1 Oct, 08:20	TRX-241001-S008	HAD05	Madina + Circle	€5.50	€6.00 rounded up	—
1 Oct, 08:20	TRX-241001-S007	HAD05	Madina + Lagon	€2.50	€3.00 rounded up	—
1 Oct, 08:20	TRX-241001-S006	CIR02	Odokor + Kassa	€6.50	€7.00 rounded up	—

Disputes

Every report with its evidence (trip, vehicle, conductor, stop history). The union sets a status: open, investigating, resolved or rejected.

FILED	TYPE	STATUS	DETAILS	ACTION
1 Oct, 10:13	Accident	Investigating	Minor collision at Kaneshie, one passenger hurt View evidence	Investigating
1 Oct, 10:13	Carless Driving	Open	Speeding and overtaking on the shoulder View evidence	open
1 Oct, 09:45	Overcharge	Open	Asked for GH¢2 instead of 1.50 View evidence	open
1 Oct, 09:45	Overcharge	Open	Asked for GH¢2 for a \$6 ride View evidence	open
1 Oct, 08:20	Overcharge	Open	I was asked for GH¢10 instead of the official GH¢7 View evidence	open
1 Oct, 08:20	Route Deviation	Open	Trotro skipped Odokor and went via the highway View evidence	open

Ratings

Average driver and conductor rating per vehicle, and the latest passenger comments. A rating below 3.0 raises a flag in Compliance.

VEHICLE	DRIVER	DRIVER RATING	CONDUCTOR RATING	RATINGS
CIR01	Kwame Mensah	4.67	4.33	3
HAD05	Kwabena Darko	4.33	4.67	3
CIR02	Kofi Asante	3	3.33	3

WHEN	VEHICLE	DRIVER	CONDUCTOR	COMMENT
1 Oct, 08:20	CIR02	2	2	Conductor asked for extra money.

Stop suggestions and USSD tester

Stop suggestions lists where passengers say they really get off, by frequency. **Add to route** puts a stop on the route (name, which stop it follows, fare between its neighbours, optional coordinates) and every passenger sees it from their next scan. The **USSD tester** lets staff try the *123# menu exactly as a handset would, which is useful for training.

7 · RULES

The rules the system enforces

These are the behaviours built into the server. Items marked **Union can change** are settings GPRTU controls; the rest are fixed by design.

RULE	WHAT HAPPENS	WHO DECIDES
Price is set by the server	The app cannot choose or change a price. A payment with the wrong amount is refused.	Fixed
Payment first	A trip is created only after MoMo confirms the payment.	Fixed
Rounding	Prices round up to a step. Today: whole cedis. A short hop of 50p would cost ₵1.00 under that rule.	Union can change (1, 0.50, 0.10, 0.05)
Fare changes	A new table takes effect at its effective time; the latest published table wins.	Union
Short-hop fares	A specific stop-to-stop fare wins over the usual difference of two stop fares.	Union
Overstay	If the trotro passes the passenger's stop, they may pay the difference or get off; ignored for 60 seconds, the extra fare is charged.	Fixed (60 s)
Registered vehicles only	Unknown codes cannot be paid; suspended vehicles cannot be paid.	Union (register, suspensions)
Conductor PIN	4 digits; 5 wrong attempts lock for 15 minutes; sessions last 12 hours; only the union resets a PIN.	Fixed (union resets)
Compliance flags	Fare >5% over; rating <3.0 (3+ ratings); 3+ overcharge reports in 30 days = Warning; 2 warnings in 90 days = suspension recommended.	Thresholds can be tuned
Direction and route changes	Only when no paid passengers are on board; always logged; more than 2 route switches a day flagged.	Fixed (guard); union reviews
Terminal fee	GH₵0.10 per trip booked to the route's starting terminal; never charged to the passenger.	Union can change
Statements	Only with recorded owner consent; app fares only; period and history required are set by the union.	Union / lender
Live alerts	Accident and careless-driving reports are urgent, show location, and can be answered in real time.	Fixed

7 · RULES

Where it could go wrong, and what stops it

The aim is to remove work, not create it. These are the weak points we looked for, with what the system does about each. Where GPRTU must decide something, it is marked.

WHAT COULD GO WRONG	WHAT THE SYSTEM DOES	GPRTU DECIDES
Passenger says they got on later than they did (to pay less)	The app finds the boarding stop from their location. At payment the phone location is compared with the stop they chose; two or more stops apart is flagged "boarding check" on the Transactions page. Nothing is refused.	What to do with repeat flags
Location is off, weak, or the phone has no GPS	Nothing is refused. The passenger picks where they got on, the timetable ETA counts down, and "I got off" is always there.	—
Fake location	A passenger's location only moves their own trip. The vehicle moves only when two different passengers' phones agree (or the conductor's phone does), never backward. Any extra charge still needs that passenger's own MoMo PIN.	—
Passenger gets off early and walks away	An extra-charge request starts only when the vehicle has gone past their stop and their own phone is still travelling with it. A passenger who left the vehicle is never asked to pay more, and even then the passenger approves any extra charge with their own MoMo PIN. Trips close by themselves ten minutes after arriving, or four hours after starting, so nobody stays on the conductor's list by mistake.	—
The stop is not on the list	Destination: the passenger types a note and pays to the stop before it. Boarding: they pick the stop just before. Both appear on the Stops page, where GPRTU can add the stop to the route in one dialog; the fare must sit between its neighbours.	Which new stops to add, and their fares
Conductor cannot read, or forgets	Conductors mark nothing. The board updates by itself. The only actions are Verify QR, Add unpaid, and a big Turn round button.	—
Trotro is running the wrong way	When someone scans while standing at the end of the line and nobody is on board, the direction flips automatically and is logged like a conductor's change. The Turn round button remains.	Whether to keep automatic turn-round
A stop has no coordinates	It still works for choosing and paying; it just cannot be detected by location. Add coordinates when adding or correcting stops.	Who maintains stop coordinates
Battery and data	The phone reports its location every ten seconds only while a paid trip is on, and only while the app is open. Scanning asks permission once.	—
Privacy	Raw coordinates are not stored. The system keeps stop names, the boarding check result, and how far from the stop a passenger was when they got off.	Data retention rules

8 · DATA

Data, privacy and security

The system collects what it needs to run the service and to protect passengers and the union, and nothing more. This page is a summary for decision-making, not legal advice.

WHO	WHAT IS STORED	WHO CAN SEE IT
Passenger	An anonymous device identifier (no name or account needed); trips and payments; the MoMo number used to pay; ratings; reports. Location only when the passenger allows it: when getting off, for the live ETA on their phone, and in an accident or careless-driving alert.	The passenger; union staff for trips, payments and reports. Passengers are not named on the dashboard.
Conductor	Vehicle code, name, last sign-in. The PIN is stored only as a protected hash, never as the PIN itself.	Union staff (status only, never the PIN).
Driver / owner	Name, phone number (for SMS), vehicle, ratings, warnings, suspensions, income statements (only with consent).	Union staff.
Union staff	Actions are recorded (warnings, suspensions, fare tables, replies).	Union leadership.

Built-in protections

- Prices and payments are decided on the server, not the phone.
- Conductor PINs and session tokens are stored as hashes.
- A report's evidence is captured by the server when it is filed.
- Income statements need the owner's consent.
- Passengers can delete their account and trip data.
- Dashboard access needs the union key.

For GPRTU to decide

- **Staff access.** Today the dashboard uses one shared union key. Individual logins with roles (for example terminal chairmen seeing only their terminal) are not built yet and are recommended before launch.
- **Legal.** Take advice on obligations under Ghana's Data Protection Act, 2012 (Act 843), data retention, and sharing statements with banks.
- **Hosting and backups.** Choose where the server runs and how data is backed up.
- **Consent wording** for owners, and a complaints route for drivers.

9 · READINESS

What is live, simulated or still needed

An honest check, so nothing is promised that is not yet true.

AREA	STATUS	DETAIL
Scan, choose, pay, ride, rate	Working	Complete flow, including mid-route boarding, short hops, overstay and arrival.
Union dashboard (all 13 sections)	Working	Compliance, fares, terminals, statements, registration, alerts and the rest.
Safety alerts and live reply	Working	Two-way, near real time. Phone-call buttons use Ghana's emergency numbers (112, 193, 191).
Direction and route changes	Working	Guarded and logged.
Conductor Big Board, head count, Verify QR	Working	Tested in the demo and against the live server.
Location-based boarding and trip progress	Test on phones	Built and tested with simulated locations. It needs trying on real phones along a real route before launch; without location everything still works by hand.
Adding unlisted stops (dashboard)	Working	GPRTU adds a stop to a route; the fare must sit between its neighbours.
Mobile-money payments	Simulated	Passengers can choose MTN MoMo, Telecel Cash or AT Money. In the demo all three auto-approve after 3 seconds and no money moves. Real payments need a collections account: MTN is wired directly; Telecel Cash and AT Money switch on once a provider that covers them is connected.
SMS (warnings, fare notices)	Simulated	Logged but not sent. Needs an SMS provider account and a sender name.
Push notifications on closed app	Needs setup	Needs a Firebase project connected to the app build.
USSD (*123#)	Needs setup	The menu is built and testable. A USSD operator and short code must be arranged.
Vehicle, conductor and driver register	Sample data	Five demo vehicles on three routes, with placeholder phone numbers. GPRTU's real register must be loaded.
Official routes, stops and fares	Sample data	Circle-Kasoa, Madina-Accra Central, Tema have sample stops and fares. GPRTU's official table replaces them.
Conductor earnings and bonus figures	Partly sample	Designed, but not yet fed from live payments.
Individual staff logins with roles	Not built	The dashboard uses one shared key today.
Passenger accounts by phone number (OTP)	Not built	Passengers are anonymous guests today.
Bank credit partnership, referral fees	Not built	Statements are ready for a lender; no bank integration.
Hosting	Not done	The server has been run for development only. It must be hosted and monitored.
Translations	First draft	Twi, Ewe and the others need native-speaker review. Conductor screens are English only.

10 · LAUNCH

Launch checklist and decisions

What GPRTU provides

- **Mobile-money collections accounts** (MTN, and Telecel Cash / AT Money through a provider) and API credentials, in the union's name.
- **SMS account** and an approved sender name (for example "GPRTU").
- **Firestore project** for push notifications on Android.
- **USSD short code** and an operator, if feature-phone riders are in scope.
- **The vehicle register:** vehicle codes, driver and conductor names, driver phone numbers, route assignments.
- **Official routes, stops (with locations) and the current fare table.**
- **Terminal list** and chairmen who will use the dashboard.
- **A host** for the server (and a domain name), plus backups and a technical contact.
- **QR stickers** printed per vehicle (the app produces them).
- **A support contact** for passengers, drivers and conductors.

Decisions for GPRTU

- **Rounding:** whole cedis, or exact to 10 pesewas for short hops?
- **Warning rules:** keep 5% over official, rating below 3.0, two warnings in 90 days?
- **Suspension:** usual length, who may order one, and an appeal route.
- **Terminal fee:** keep GH¢0.10 per trip? Who receives it?
- **Route changes:** allowed freely (logged), or should the union approve them?
- **Statements:** how many weeks or months a lender needs; consent wording.
- **Access:** who gets dashboard access, and what each role may see.
- **Languages:** who reviews the Twi and Ewe wording?

A suggested way to start

These are suggestions to discuss, not commitments.

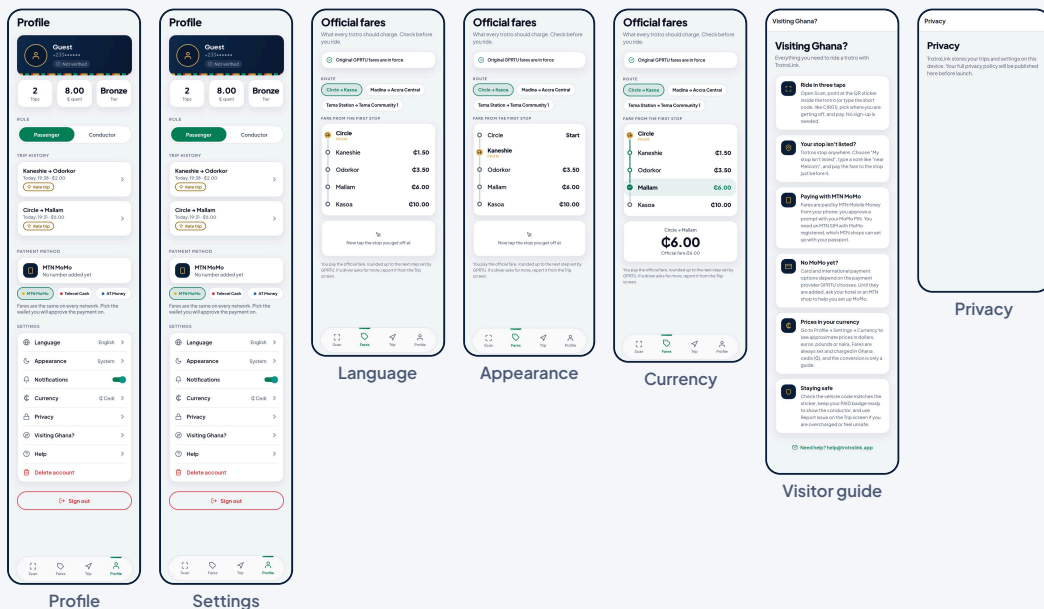
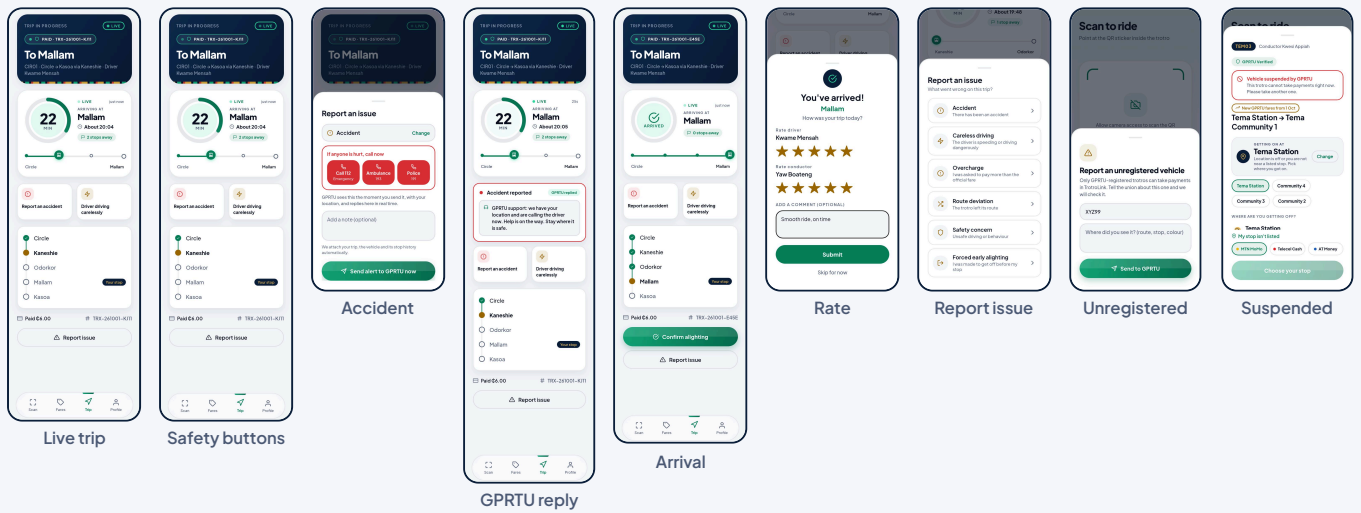
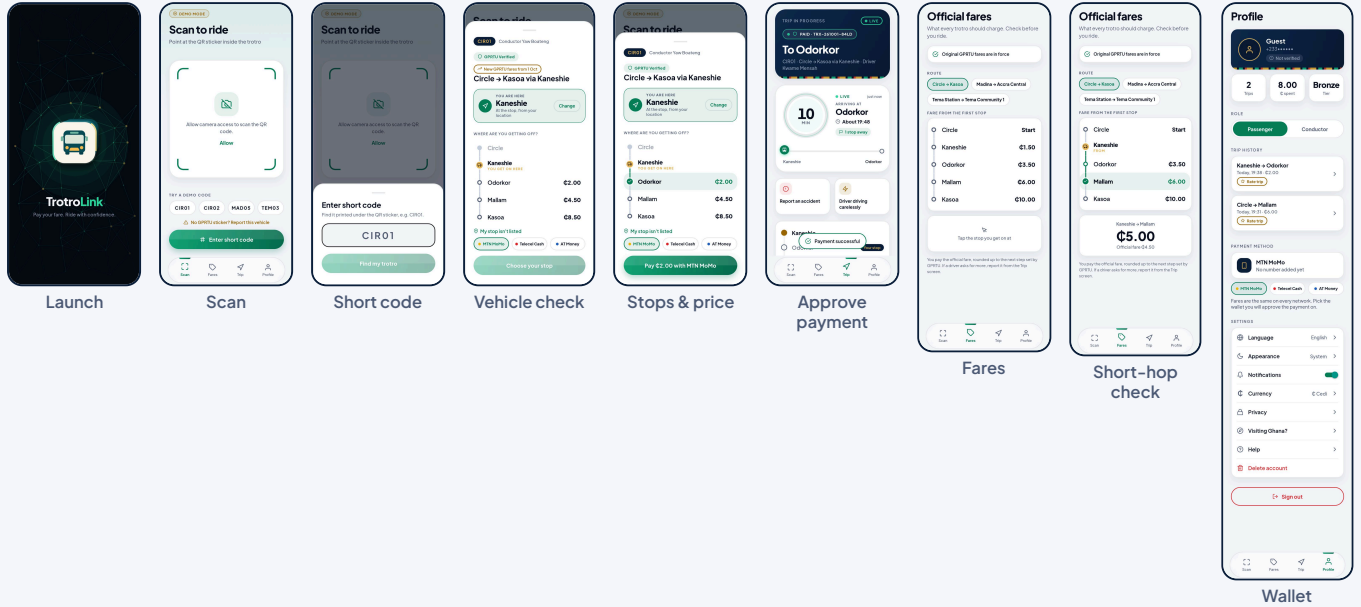
- **1 · Prepare.** Load the real register and fares; connect MoMo and SMS; host the server.
- **2 · Pilot.** One route and terminal, a small group of vehicles, trained conductors, stickers on dashboards.
- **3 · Review.** Weekly: fares paid, reports, ratings, alerts, and passenger and driver feedback.
- **4 · Expand.** More terminals; switch on USSD; start statements as history builds.

APPENDIX A

Every screen at a glance

The phone screens, grouped by user. The dashboard sections are listed in section 6.

Passenger (25)



APPENDIX B

Glossary

Plain-English meanings of the terms used in this guide.

TERM	MEANING
API / server	The central computer program that every phone and the dashboard talk to. It holds the rules and the records.
Compliance flag	A warning marker the system puts on a vehicle when it breaks one of the union's rules, such as fares too high.
Dashboard	The web page union staff use to watch and manage the system.
Fare table	The list of official fares between stops, with an effective date. The system always uses the latest one that has taken effect.
GPRTU Verified	The badge a passenger sees when a vehicle is on the union's register.
Live alert	An accident or careless-driving report that appears on the dashboard immediately, with a location.
Mobile money (MoMo)	MTN MoMo, Telecel Cash or AT Money: the wallets passengers pay from.
Overstay	When the trotro goes past the stop the passenger paid for.
QR sticker / short code	The picture the passenger scans, and the short code under it (for example CIRO1), which identifies the vehicle.
Rounding step	How much a passenger's price is rounded up (whole cedis, or finer such as 10 pesewas).
Short hop	A short trip between two stops, which the union can price exactly.
Simulator	A pretend version of an outside service (such as MoMo or SMS) used for demonstrations. Nothing real happens.
Statement	A report of a vehicle's fares collected through the app, by week or month, shared with the owner's consent.
Suspension	The union stopping a vehicle from taking app payments for a number of days.
Terminal fee	A small fee booked per trip to the terminal where the route starts. Not charged to the passenger.
USSD	A menu you open by dialling a code like *123# on any phone, with no internet or app.